Julio Alfredo Ballinas García. 24/11/2025

En la región I, $V(x)=0$: $u''_I(x) + 2iKu'_I(x) - [K^2-\alpha^2]\,u_I(x) = 0$

En la región II, $V(x)=V_0$: $u''_{II}(x) + 2iKu'_{II}(x) - [K^2-\beta^2]\,u_{II}(x) = 0$

donde se ha definido $\beta = \sqrt{\dfrac{2m(\varepsilon - V_0)}{\hbar^2}}$ y $\alpha = \sqrt{\dfrac{2m\varepsilon}{\hbar^2}}$

β es imaginario puro si $0<\varepsilon<V_0$

se proponen las siguientes soluciones

$$u_I(x) = Ae^{i(\alpha-K)x} + Be^{-i(\alpha+K)x} \qquad 0<x<a$$

$$u_{II}(x) = Ce^{i(\beta-K)x} + De^{-i(\beta+K)x} \qquad -b<x<0$$

usamos condiciones de continuidad para la función de onda y su derivada ($u(x)$ y $u'(x)$) en los puntos apropiados, tenemos...

$$u_I(x)\Big|_0 = u_{II}(x)\Big|_0 \qquad u_I(x)\Big|_a = u_{II}(x)\Big|_{-b}$$

$$u'_I(x)\Big|_0 = u'_{II}(x)\Big|_0 \qquad u'_I(x)\Big|_a = u'_{II}\Big|_{-b}$$

se obtiene un sistema de 4 ecuaciones y 4 incógnitas cuyo determinante está descrito como:

$$0 = \begin{vmatrix} 1 & 1 & -1 & -1 \\ \alpha - K & -(\alpha+K) & -(\beta-K) & (\beta+K) \\ e^{i(\alpha-K)a} & e^{i(\alpha+K)a} & -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \\ (\alpha-K)e^{i(\alpha-K)a} & -(\alpha+K)e^{-i(\alpha+K)a} & -(\beta-K)e^{i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix}$$

Tenemos resolviendo

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

Tenemos

$$1 \begin{vmatrix} -(\alpha+K) & -(\beta-K) & (\beta+K) \\ e^{-i(\alpha+K)a} & -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \\ -(\alpha+K)e^{-i(\alpha+K)a} & -(\beta-K)e^{-i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix}$$

$$-(\alpha-K) \begin{vmatrix} 1 & -1 & -1 \\ e^{-i(\alpha+K)a} & -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \\ -(\alpha+K)e^{-i(\alpha+K)a} & -(\beta-K)e^{-i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix}$$

$$+ e^{i(\alpha-K)a} \begin{vmatrix} 1 & -1 & -1 \\ -(\alpha+K) & -(\beta-K) & (\beta+K) \\ -(\alpha+K)e^{-i(\alpha+K)a} & -(\beta-K)e^{-i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix}$$

$$-(\alpha-K)e^{i(\alpha-K)a} \begin{vmatrix} 1 & -1 & -1 \\ -(\alpha+K) & -(\beta-K) & (\beta+K) \\ e^{-i(\alpha+K)a} & -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \end{vmatrix}$$

Tenemos primer término

$$1\begin{vmatrix} -(\alpha+K) & -(\beta-K) & (\beta+K) \\ e^{-i(\alpha+K)a} & -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \\ -(\alpha+K)e^{-i(\alpha+K)a} & -(\beta-K)e^{-i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix} =$$

$$= -(\alpha+K)\begin{vmatrix} -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \\ -(\beta-K)e^{-i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix} - e^{i(\alpha+K)a} \cdot$$

$$\begin{vmatrix} -(\beta-K) & (\beta+K) \\ -(\beta-K)e^{-i(\beta-K)b} & (\beta+K)e^{i(\beta+K)b} \end{vmatrix} - (\alpha+K)e^{-i(\alpha+k)a} \cdot$$

$$\begin{vmatrix} -(\beta-K) & (\beta+K) \\ -e^{-i(\beta-K)b} & -e^{i(\beta+K)b} \end{vmatrix}$$

Segundo término: